Date:
Exp No.10

Aim:Implementation of Disk Scheduling using FCFS,SCAN,C-SCAN,LOOK,C-LOOK and SSTF algorithm

1.FCFS(FIRST COME FIRST SERVE)

1.Write the c program OS_exp10_1.c

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
   int n, i, head, total = 0;

   printf("Enter number of requests: ");
   scanf("%d", &n);

   int req[n];

   printf("Enter the request sequence:\n");
   for(i = 0; i < n; i++)
   {
      scanf("%d", &req[i]);
   }

   printf("Enter initial head position: ");
   scanf("%d", &head);

   for(i = 0; i < n; i++)
   {
      total = total + abs(head - req[i]);
      head = req[i];
   }

   printf("Total head movement = %d\n", total);

   return 0;
}
```

2.Compile and Run

```
gcc OS_exp10_1.c -o  OS_exp10_1
./ OS_exp10_1
```

3.Expected output

```
Enter number of requests: 8
Enter the request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Total head movement = 640
```

2.SCAN

1.Write the c program OS_exp10_2.c

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[50], n, head, total = 0, i, j, temp, dir;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);

    printf("Enter initial head position: ");
    scanf("%d",&head);

    printf("Enter direction (0=left,1=right): ");
    scanf("%d",&dir);

    /* sorting */
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(req[i] > req[j])
            {
                temp=req[i];
                req[i]=req[j];
                req[j]=temp;
            }

    int index;
    for(i=0;i<n;i++)
        if(head < req[i])
        {
            index=i;
            break;
        }

    if(dir==1)   // right direction
    {
        for(i=index;i<n;i++)
        {
            total += abs(head-req[i]);
            head=req[i];
        }

        total += abs(head-199);
        head=199;
```

```
        for(i=index-1;i>=0;i--)
        {
            total += abs(head-req[i]);
            head=req[i];
        }
    }
    else     // left direction
    {
        for(i=index-1;i>=0;i--)
        {
            total += abs(head-req[i]);
            head=req[i];
        }

        total += abs(head-0);
        head=0;

        for(i=index;i<n;i++)
        {
            total += abs(head-req[i]);
            head=req[i];
        }
    }

    printf("Total head movement = %d\n",total);

    return 0;
}
```

2.Compile and Run

```
gcc OS_exp10_2.c -o  OS_exp10_2
./ OS_exp10_2
```

3.Expected output

```
Enter number of requests: 8
Enter request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter direction (0=left,1=right): 0
Total head movement = 236
```

3.C-SCAN

1.Write the c program OS_exp10_3.c

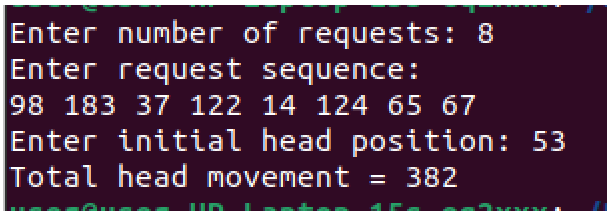
```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[50], n, head, total = 0, i, j, temp;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);

    printf("Enter initial head position: ");
    scanf("%d",&head);

    /* Sorting the request queue */
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(req[i] > req[j])
            {
                temp=req[i];
                req[i]=req[j];
                req[j]=temp;
            }

    int index;
    for(i=0;i<n;i++)
        if(head < req[i])
        {
            index=i;
            break;
        }

    /* Move towards right */
    for(i=index;i<n;i++)
    {
        total += abs(head - req[i]);
        head = req[i];
    }

    /* Move to end of disk */
    total += abs(head - 199);
    head = 199;

    /* Jump to beginning */
```

```
    total += abs(head - 0);
    head = 0;

    /* Continue servicing */
    for(i=0;i<index;i++)
    {
        total += abs(head - req[i]);
        head = req[i];
    }

    printf("Total head movement = %d\n", total);

    return 0;
}
```

2.Compile and Run

```
gcc OS_exp10_3.c -o  OS_exp10_3
./ OS_exp10_3
```

3.Expected output



4.LOOK

1.Write the c program OS_exp10_4.c

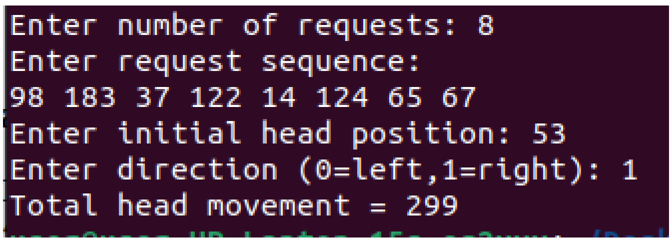
```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[50], n, head, total = 0, i, j, temp, dir;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);

    printf("Enter initial head position: ");
    scanf("%d",&head);

    printf("Enter direction (0=left,1=right): ");
    scanf("%d",&dir);

    /* Sort requests */
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(req[i] > req[j])
            {
                temp=req[i];
                req[i]=req[j];
                req[j]=temp;
            }

    int index;
    for(i=0;i<n;i++)
        if(head < req[i])
        {
            index=i;
            break;
        }

    if(dir==1)   // move right
    {
        for(i=index;i<n;i++)
        {
            total += abs(head-req[i]);
            head=req[i];
        }

        for(i=index-1;i>=0;i--)
```

```
        {
            total += abs(head-req[i]);
            head=req[i];
        }
    }
    else        // move left
    {
        for(i=index-1;i>=0;i--)
        {
            total += abs(head-req[i]);
            head=req[i];
        }

        for(i=index;i<n;i++)
        {
            total += abs(head-req[i]);
            head=req[i];
        }
    }

    printf("Total head movement = %d\n",total);

    return 0;
}
```

2.Compile and Run

gcc OS_exp10_4.c -o OS_exp10_4
./ OS_exp10_4

3.Expected output



```
Enter number of requests: 8
Enter request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter direction (0=left,1=right): 1
Total head movement = 299
```

5.C-LOOK

1.Write the c program OS_exp10_5.c

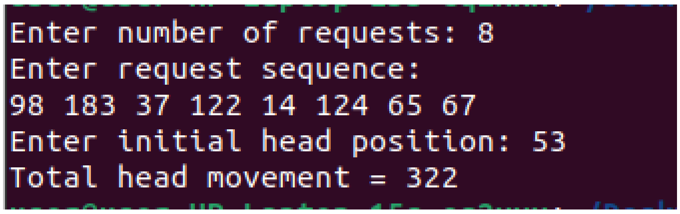
```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[50], n, head, total = 0, i, j, temp;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);

    printf("Enter initial head position: ");
    scanf("%d",&head);

    /* Sort the request queue */
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(req[i] > req[j])
            {
                temp=req[i];
                req[i]=req[j];
                req[j]=temp;
            }

    int index;
    for(i=0;i<n;i++)
        if(head < req[i])
        {
            index=i;
            break;
        }

    /* Move towards right */
    for(i=index;i<n;i++)
    {
        total += abs(head-req[i]);
        head = req[i];
    }

    /* Jump to first request */
    if(index > 0)
    {
        total += abs(head - req[0]);
        head = req[0];
    }
```

```
    /* Continue servicing */
    for(i=1;i<index;i++)
    {
        total += abs(head - req[i]);
        head = req[i];
    }

    printf("Total head movement = %d\n", total);

    return 0;
}
```

2.Compile and Run

```
gcc OS_exp10_5.c -o  OS_exp10_5
./ OS_exp10_5
```

3.Expected output



```
Enter number of requests: 8
Enter request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Total head movement = 322
```

6.SSTF(SHORTEST SEEK TIME FIRST)

1.Write the c program OS_exp10_6.c

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[50], visited[50];
    int n, head, total = 0, i, j, index, min;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&req[i]);
        visited[i] = 0;
    }

    printf("Enter initial head position: ");
    scanf("%d",&head);

    for(i=0;i<n;i++)
    {
        min = 1000;

        for(j=0;j<n;j++)
        {
            if(visited[j] == 0)
            {
                int dist = abs(head - req[j]);

                if(dist < min)
                {
                    min = dist;
                    index = j;
                }
            }
        }

        total += min;
        head = req[index];
        visited[index] = 1;
    }

    printf("Total head movement = %d\n", total);

    return 0;
```

}

2.Compile and Run

gcc OS_exp10_6.c -o OS_exp10_6
./ OS_exp10_6

3.Expected output

```
Enter number of requests: 8
Enter request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Total head movement = 236
```

Result

Thus the Disk Scheduling algorithms FCFS, SSTF, SCAN, C-SCAN, LOOK and C-LOOK were implemented successfully